

FTP Guide for Thaghrah

Description

FTP (File Transfer Protocol) is a standard network protocol used to transfer files between a client and a server on a computer network. It is one of the oldest protocols still in use today and operates on the **Application Layer** of the TCP/IP model.

FTP is a **client-server** protocol that uses **two separate TCP connections**:

- **Control Connection** (usually port 21): For sending commands and receiving responses.
- **Data Connection** (port 20 in active mode, or a dynamic port in passive mode): For actually transferring files and directory listings.

FTP is **stateful** — it maintains a session with the server until the user logs out or the connection is closed. While powerful, traditional FTP transmits data (including usernames and passwords) in **plain text**, making it insecure for modern use.

Key Components of FTP

1. Connection Modes

- **Active Mode**: The client tells the server which port to connect to for data transfer (server initiates the data connection).
- **Passive Mode (PASV)**: The server opens a port and tells the client to connect to it (more firewall-friendly, most commonly used today).

2. Authentication

- Usually requires a username and password.
- Supports **anonymous FTP** (common username: anonymous or ftp, often with email as password).

3. Common FTP Commands

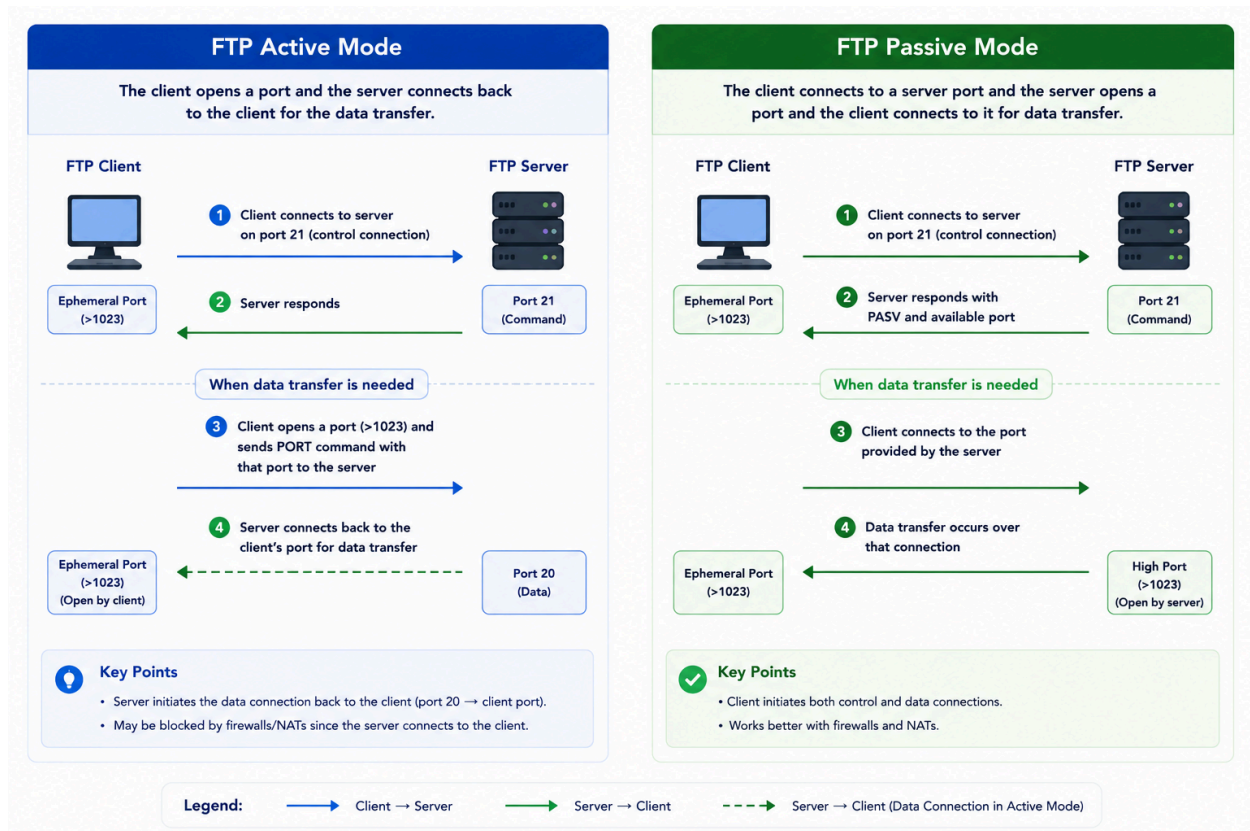
- USER — Send username
- PASS — Send password
- LIST / NLST — List files and directories
- RETR — Download (Retrieve) a file
- STOR — Upload (Store) a file
- PWD — Print Working Directory
- CWD — Change Working Directory
- QUIT — Logout and close connection

4. File Transfer Types

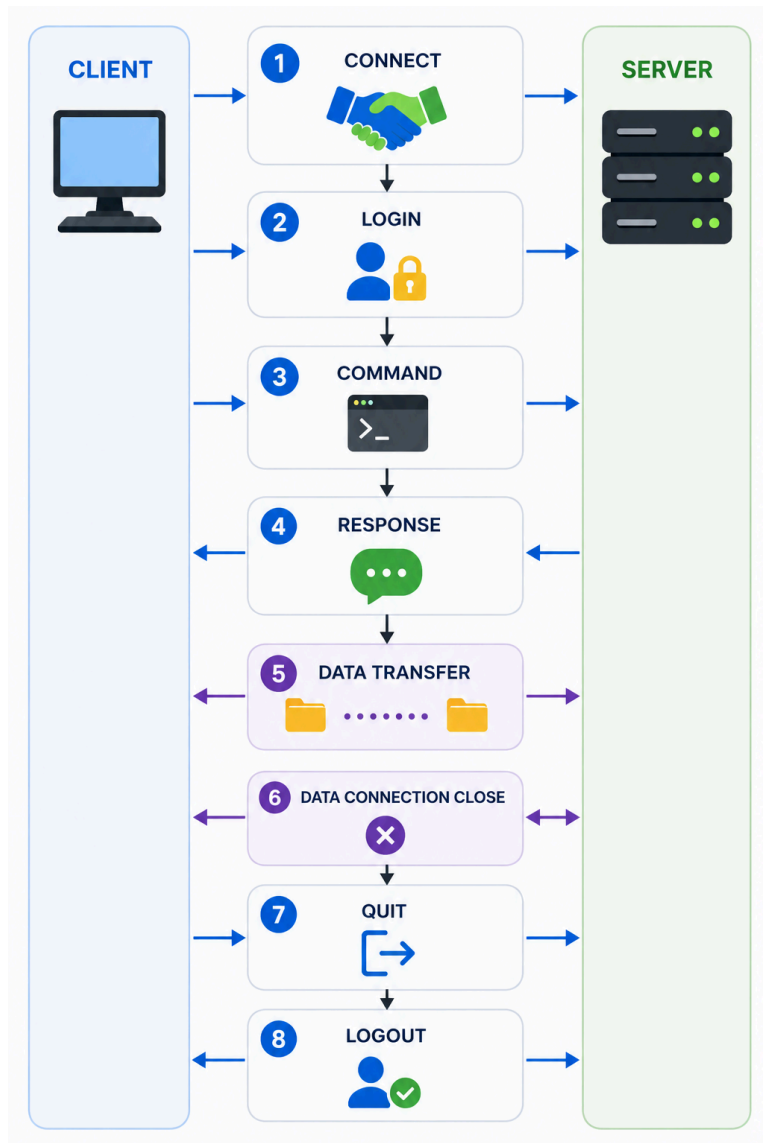
- **ASCII Mode:** For text files
- **Binary Mode:** For images, executables, archives, etc.

Visualizations

FTP Active vs Passive Mode:



FTP Command Flow:



FTP - File Transfer Protocol Animated

<https://youtu.be/kKxSiJPQ2kg?si=TTS4pY7ECTTyAjFS>